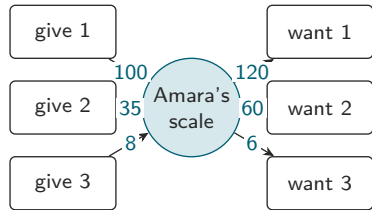


$n \times m$ rates to state
and they can contradict each
other: $r_{11} r_{22} \neq r_{12} r_{21}$



$n + m$ prices on one private scale
every rate is a ratio p_j/p_i :
no arbitrage possible

The scale is pure bookkeeping: nothing is issued, held or transferred, and each maker's scale is their own. Only ratios ever leave the maker; around a loop the scales cancel.